DHCP
(DYNAMIC HOST CONFIGURATION PROTOCOL)

Dynamic Host Configuration Protocol (DHCP) is a network protocol that automatically assigns **IP** addresses and other network configuration details to devices in a network. A **DHCP** server holds a pool of IP addresses and gives one to each device that requests it, along with the subnet mask, default gateway, and **DNS** server address.
It works through a four-step process known as **DORA**: Discover, Offer, Request, and Acknowledge. The client broadcasts a request, the server offers an available address, the client accepts it, and the server confirms the lease. The address is assigned for a limited time, called the lease time, after which it must be renewed or released.
DHCP is widely used in homes, schools, offices, and large organizations, where manually configuring each device would be time-consuming and error-prone**.**

Advantages of DHCP

▪ Automatically assigns IP addresses, which removes the need to configure each device manually.
▪ Reduces configuration errors such as duplicate IP addresses and wrong subnet masks.
▪ Makes network management easier because all addressing is controlled from a central server.
▪ Allows efficient use of IP addresses, since unused addresses return to the pool when leases expire.
▪ Easy to expand because new devices receive their configuration automatically when they connect.
▪ Supports mobile devices that move between networks, as they receive suitable settings automatically.

Disadvantages of DHCP

▪ The DHCP server is a single point of failure, so if it fails, new devices cannot obtain IP addresses.
▪ Devices receive changing IP addresses unless reservations are configured, which makes some services harder to manage.
▪ Unauthorized (rogue) DHCP servers can give wrong settings or be used for attacks.
▪ Requires proper configuration of address pools, lease time, and excluded addresses.
▪ Servers and network devices still need static addresses, so DHCP cannot replace all manual configuration.
▪ If the address pool is too small, new devices cannot join the network once it is exhausted.

Simulation / Design in Cisco Packet Tracer

This section presents the **DHCP** network designed and simulated in Cisco Packet Tracer. The network consists of a **DHCP** server and multiple PCs connected to a central switch. The server automatically assigns IP addresses, subnet masks, default gateway, and **DNS** server details to the PCs, demonstrating the working of DHCP.

Network Diagram

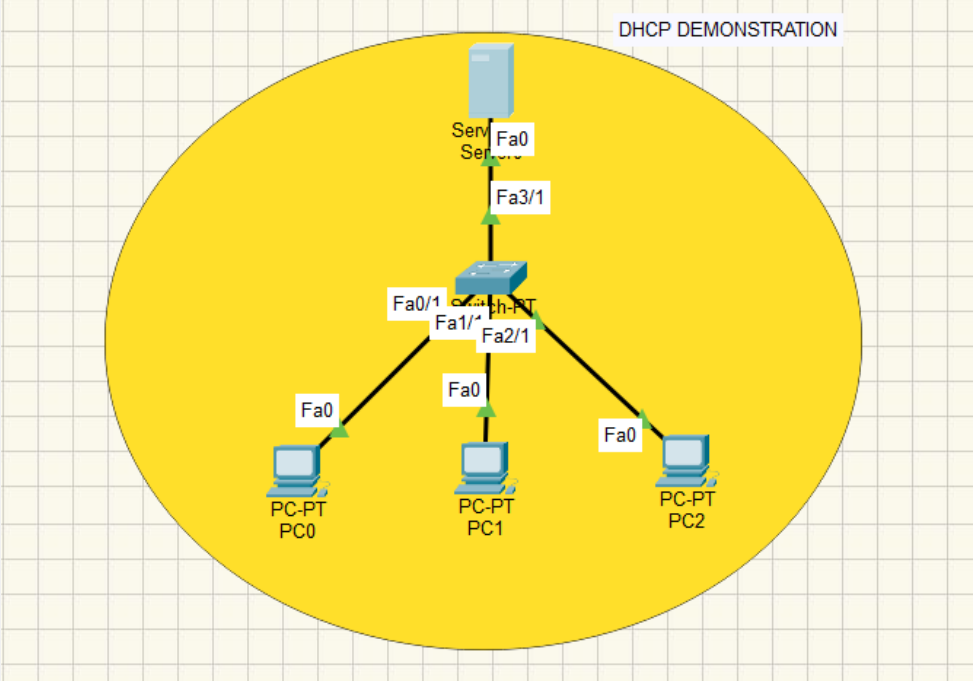


Figure 1: DHCP network layout in Cisco Packet Tracer

Device Configuration

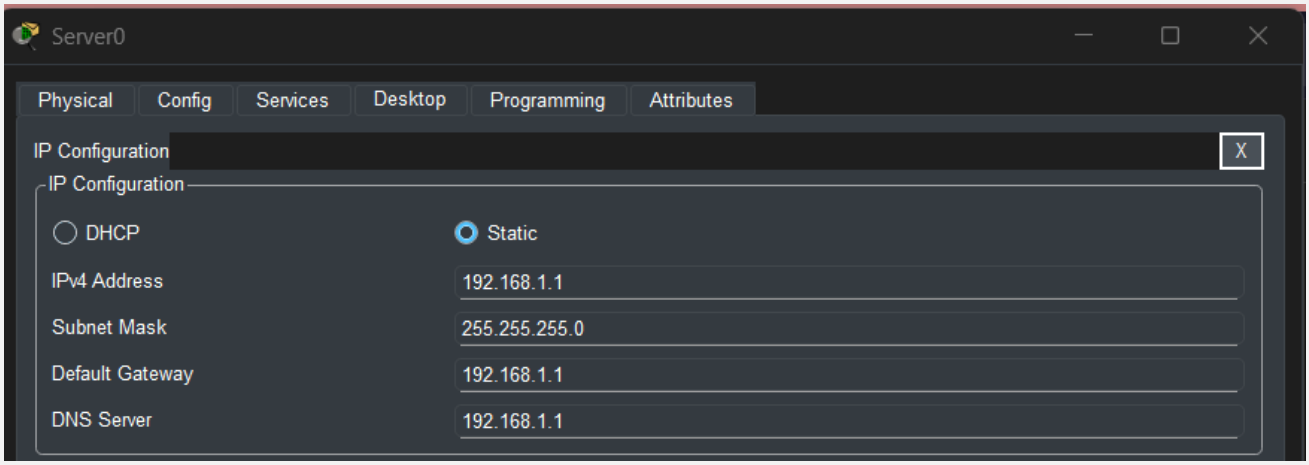


Figure 2.1: Static IP configuration of the DHCP server (192.168.1.1)

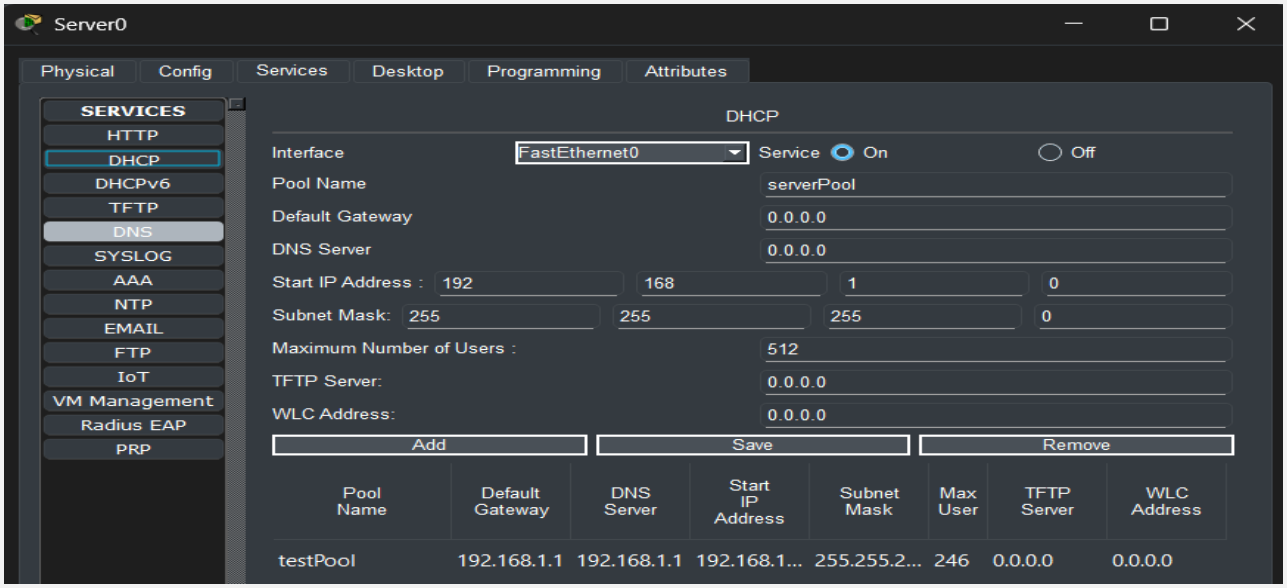


Figure 2.2: DHCP service configuration window of the server

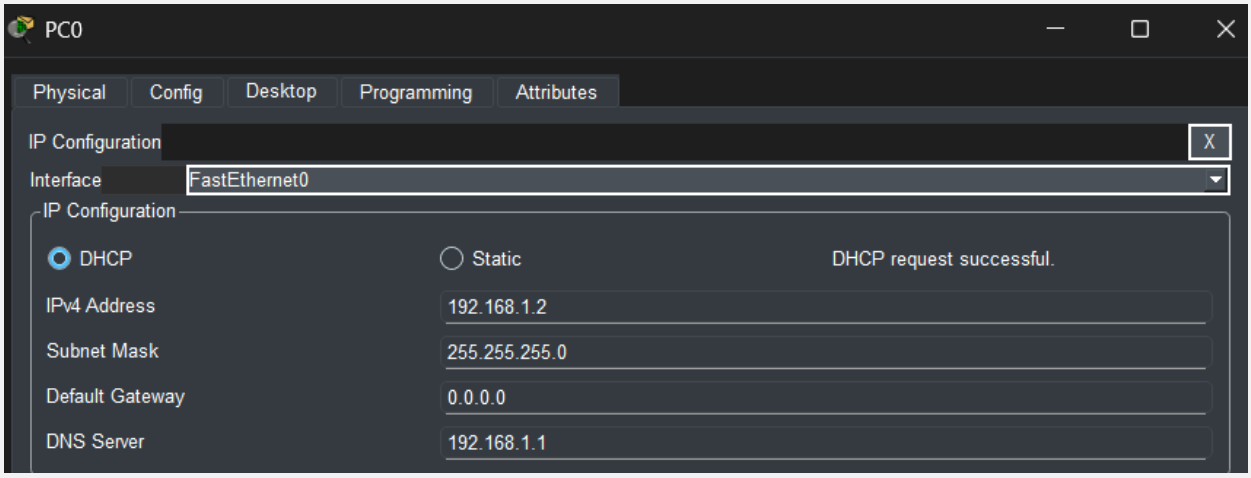


Figure 2.3: IP address automatically assigned to PC0

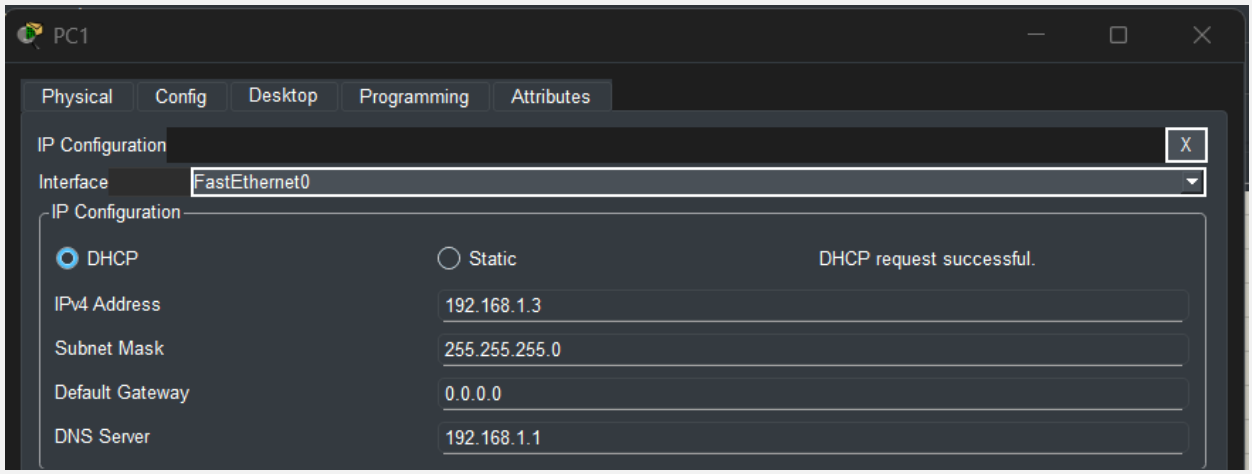


Figure 2.3: IP address automatically assigned to PC1

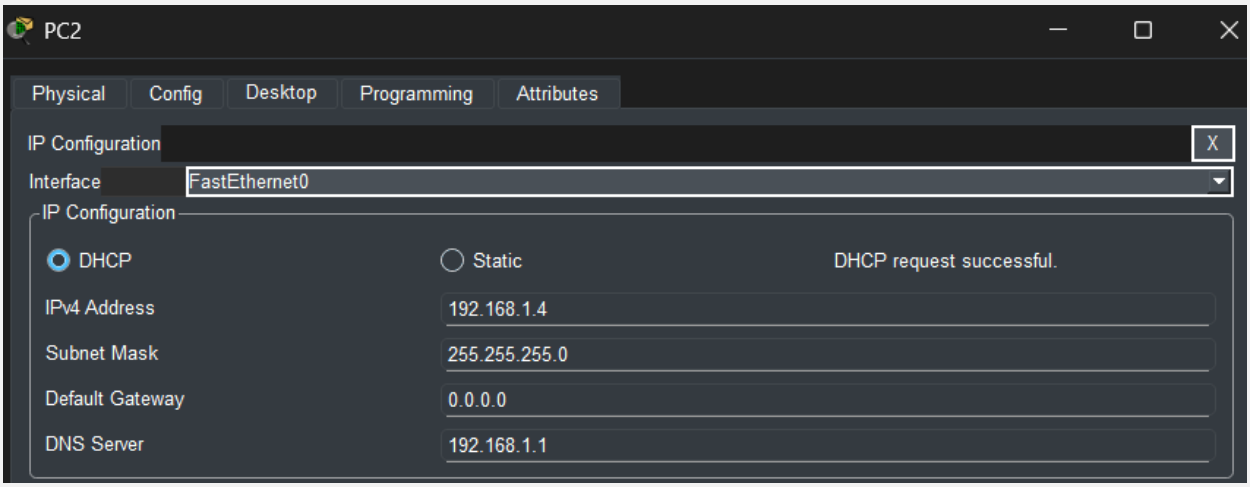


Figure 2.3: IP address automatically assigned to PC2

Connectivity Test (DHCP/ Simulation Result)

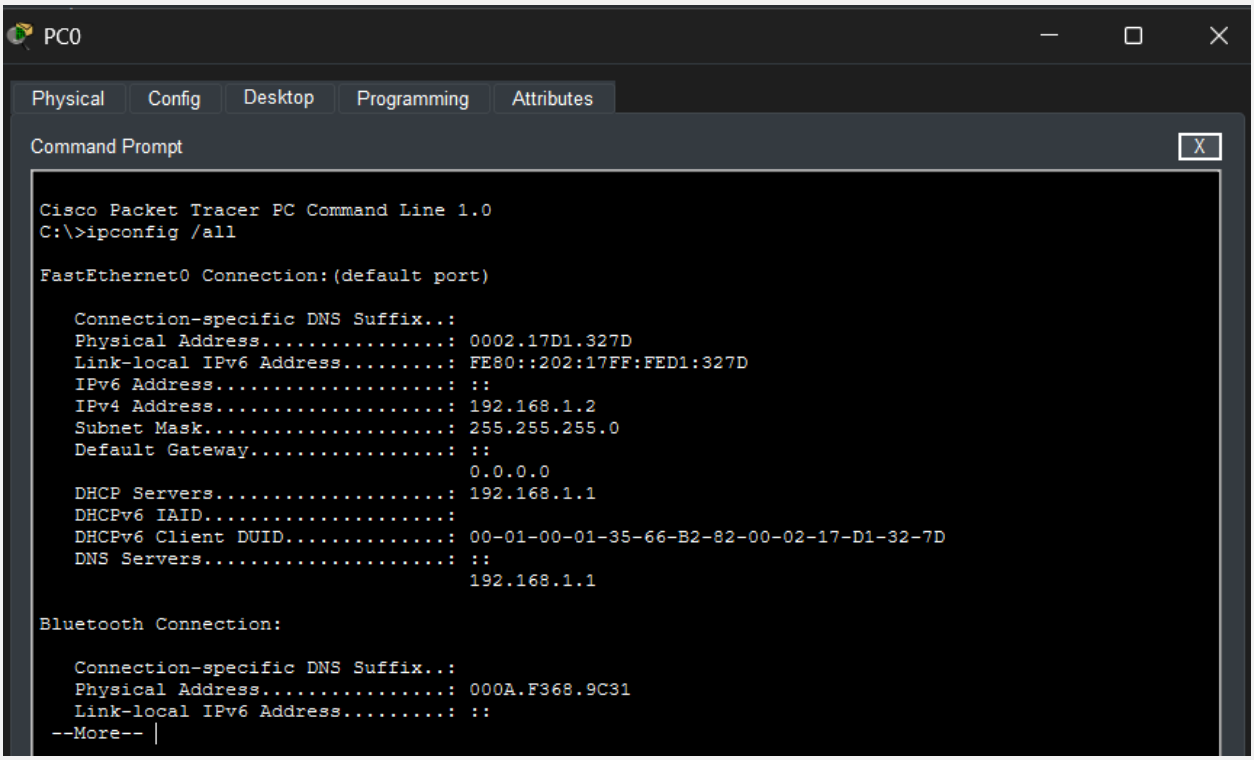


Figure 3.1: ipconfig output showing the IP address received from the DHCP server

Process of Designing the Network in Packet Tracer

- Open Cisco Packet Tracer and create a new blank project to begin designing the DHCP network.
- Build a minimal topology by placing one 2960 switch at the center of the workspace. From the End Devices category, place one Server-PT, which will act as the DHCP server, and three PC-PTs, named PC0, PC1, and PC2, which will act as DHCP clients.
- Select Copper Straight-Through cable from the Connections category and connect the Server and each PC individually to the switch. Wait until the link lights turn green, which shows that the physical connections are active.
- Give the server a static IP address, because the DHCP server must keep a fixed address. Open Server → Desktop → IP Configuration, select Static, and enter the values shown below.

Field	Value
IPv4 Address	192.168.1.1
Subnet Mask	255.255.255.0
Default Gateway	192.168.1.1
DNS Server	192.168.1.1

- The server's address must lie outside the pool created in the next step, so that it is never handed out to a client.
- Configure the DHCP service by opening Server → Services → DHCP and setting the service to On.
- Edit the default pool (serverPool) and enter the following details, then click Save.

Field	Value
Pool Name	LabPool
Default Gateway	192.168.1.1
DNS Server	192.168.1.1
Start IP Address	192.168.1.10
Subnet Mask	255.255.255.0

- With these settings, the server will hand out addresses from 192.168.1.10 to 192.168.1.59.
- Make the clients request an address. On each PC, open Desktop → IP Configuration and select the DHCP option instead of Static.
- The message "DHCP request sent" appears briefly. The PC then displays its newly assigned IP address, subnet mask, default gateway, and DNS server, confirming that the DHCP server has configured the client automatically.
- Verify the leases on each PC by opening the Command Prompt and entering the command ipconfig /all. The output shows the IP address, subnet mask, default gateway, DNS server, and the DHCP server address (192.168.1.1) for that PC.
- Compare the addresses on PC0, PC1, and PC2. Each PC receives a different address from the pool, starting from 192.168.1.10, which confirms that the server assigns unique addresses without any manual configuration.

- Optionally, test connectivity between the clients by using the ping command from one PC to the address of another PC. Successful replies confirm that the DHCP-assigned configuration is working correctly.
- Screenshots of the topology, the server's static IP configuration, the DHCP pool settings, the DHCP request on the PCs, and the ipconfig /all output are captured as evidence of the completed setup.
- Finally, the completed Packet Tracer project is saved as a .pkt file, along with the required documentation and screenshots for submission

Conclusion

The **DHCP** network was successfully designed and configured in Cisco Packet Tracer, demonstrating how IP addresses can be assigned automatically to multiple devices. The experiment involved configuring a DHCP server with an address pool, setting the PCs to obtain addresses dynamically, and verifying the assigned configuration. Connectivity was verified using the ipconfig and ping commands and Simulation Mode. This experiment reinforced the practical understanding of DHCP, the **DORA** process, and automatic network configuration.